

Haritha Hospitals

ERPNext + HRMS deployment for **Processbricks' Haritha Hospitals** project — a real hospital shift-management and HRMS rollout. Built on Frappe v16 with a custom app (haritha_hospital) that captures all customizations as portable fixtures, plus a migration playbook for replicating any env.

Owner	Venkat Narasimha (Processbricks)
Started	2026-04-01 (concept); project on this repo 2026-08-19
Stack	Frappe v16.30.0 · ERPNext v16.30.0 · HRMS v16.5.0 (pinned) · custom app haritha_hospital
Envs	pberpprod.duckdns.org (prod) · pberpdev.duckdns.org (dev) · dev-erp.duckdns.org (Venkat VPS prototyping)
Scope	Shift management + HRMS basics (employees, departments, shift types, attendance, leave)
Out of scope	Wards, beds, OTs, pharmacy, lab, billing, full Chart of Accounts, cost centers beyond the 2 created (deferred)

Project Status (2026-08-31)

- **Custom app haritha_hospital built** — captures **274 production-validated customizations** as Frappe JSON fixtures (78 Custom Fields + 189 Property Setters + 3 Print Formats + 2 Notifications + 2 Letter Heads).
- **Installed on both envs** — pberpdev (fresh, all 274 loaded) and pberpprod (idempotent re-install verified, installed_apps updated, count matches).
- **Master data migrated prod → dev** — 16 DocTypes via scripts/migrate_master_data.py. pberpdev now mirrors prod: 1 Company, 210 Employees, 37 Departments, 25 Shift Types, 8,118 Shift Assignments, 6,300 Attendance, 12,562 Employee Checkins, etc.
- **Phase 6 + Tier 6 documentation complete** — 35 docs across 9 tiers (docs/phase6/), ~15,900 lines. Tier 0-5 (foundations, schema, workflow, client, runbooks, process) + Tier 6/Tier 9 (09-compliance/, ISO 27001-aligned policies + CMM L5 maturity) + Tier 7-8 (user manuals + testing). Mermaid diagrams rendered to PNG.
- **Client demo deck + speaker script** — docs/phase6/03-client/: 11-slide deck (03.4), PPTX export (03.4.pptx, 543 KB), speaker script (03.5), 10 screenshots from pberpprod, 20 mermaid-rendered diagrams.
- **P1 outage resolved (2026-08-29 03:06 IST)** — unicorn --preload sys.path freeze after install-app; both envs restarted (~30-60s downtime each, zero data loss). Always restart backend after install-app (LEARNINGS #153).
- **Roster SPA verified rendering** — Phase 4.10/4.11 fixes held; /hr/roster shows 211 employees × 31 days without crash.

Phase	State
Phase 0 — Schema planning	☐ done
Phase 1 — Schema approval	☐ done

Phase	State
Phase 2 — Site setup (pberpprod)	☐ done
Phase 3 — Data import (24,511 records)	☐ done
Phase 3.5-3.10 — Reconcile / bulk-submit / property setters / linkage	☐ done
Phase 4 — Roster crash + Attendance HRMS-recompute	☐ done
Phase 0+ — Custom app + master data migration + outage recovery	☐ done
Phase 5 — Production readiness (DR, security, perf, UAT)	☐ skipped per Venkat
Phase 6 — Process & maturity docs	☐ done (35 docs, 9 tiers, 2026-08-29)
Phase 6 — Tier 6 Compliance/Maturity	☐ done (12 docs: ISO 27001 + CMM L5, 2026-08-29)
Client demo deck + script	☐ done (2026-08-30)

Recent Deliverables (2026-08-30)

Deliverable	Path	Notes
Client demo deck	docs/phase6/03-client/03.4-client-presentation.pptx	11 slides, 543 KB, screenshots + diagrams embedded
Speaker script	docs/phase6/03-client/03.5-speaker-script.md	Talking notes for each slide
Demo screenshots	docs/phase6/03-client/screenshots/	10 PNGs from pberpprod (login, dashboard, roster, etc.)
Diagram assets	docs/phase6/03-client/assets/	20 mermaid-rendered PNGs
Phase 6 + Tier 6 docs	docs/phase6/	35 docs, ~15,900 lines, 9 tiers

Stack

- **Frappe** v16.30.0 — foundation framework
- **ERPNext** v16.30.0 — accounting, inventory, selling, buying
- **HRMS** v16.5.0 — pinned per Lesson #44 (v16.5.1+ breaks on repost_allowed_types)
- **MariaDB** 10.x — Docker named volumes, restart-safe
- **Redis** — cache + queue broker; Socket.IO for realtime desk
- **Docker Compose** — compose-based deployment (erp-{env}-* containers on main VPS, erpdev-* on Venkat VPS)
- **nginx-proxy** — reverse proxy with TLS termination (DuckDNS + Let's Encrypt / self-signed fallback)
- **Custom app** haritha_hospital (0.0.1) — owns the 274 customizations as fixtures
- **Python 3.11** — Frappe v16 baseline

- **3 environments:** pberpdev (dev), pberpqa (QA — skipped for Haritha per Venkat), pberpprod (prod)

Repo layout

Path	Purpose
TRACKER.md	Project tracker — phase history, subagent log, decisions
README.md	This file — top-level project overview
HARITHA_HOSPITALS_GUIDE.md	(at docs/HARITHA_HOSPITALS_GUIDE.md) — comprehensive end-to-end guide (architecture, customizations, migration, ops)
INDEX.md	Curated reading index
DECISIONS.md	Decision log
WORKFLOW.md	Shift management workflow notes
PRODUCTION-READINESS-AUDIT-2026-08-21.md	Pre-rollback audit
MIGRATION-GUIDE.md	Migration playbook reference
all_schemas.csv	Schema definitions for 15 master entities (schema-only, single CSV)
masters/	Source CSVs — 19 files, ~1.77 MB, 24,758 rows (Company, Department, Designation, Employee, Shift Type, Shift Assignment, Attendance, Employee Checkin, etc.)
docs/	Documentation — HARITHA_HOSPITALS_GUIDE.md + phase6/ (Tier 0-8) + pberp-setup-plan.md
docs/phase6/	Phase 6 documentation — 35 docs, 9 tiers, ~15,900 lines (foundations, schema, workflow, client, runbooks, process, user manuals, testing, compliance)
docs/phase6/03-client/	Client demo deliverable — markdown deck (03.4), PPTX export (03.4.pptx, 543 KB), speaker script (03.5), 10 screenshots from pberpprod, 20 mermaid-rendered diagrams
docs/phase6/09-compliance/	ISO 27001 + CMM L5 docs (12 policies/maturity docs, internal best-practice reference)
tracker-phases/	Project tracker split into per-phase files (master TRACKER.md is the index)
scripts/	Utility scripts (~45 files) — migrate_master_data.py, recreate_property_setters.py, bulk_submit.py, fix_attendance_hrms_recompute.py, verify_csvs.py, update_tracker.py, etc.
fixtures/	Legacy fixtures dir (pre-custom-app)
mapping/	Data mapping rules
scout/	Source data scout reports
phase-a/	Phase A fixtures bundle (fixtures.tar.gz) + REPORT.md + logs/

Path	Purpose
audit/	Fixture audit reports (pberpprod detail + summary)
config/	cron.tab for backup cron
updates/	Phase update JSON snapshots
reports/	Generated reports
.gitignore	Standard Frappe ignores

There is no memory/ or uploads/ at this level. Daily logs and source data live in the OpenClaw workspace, not the project repo.

Customizations catalog

All captured as Frappe fixtures in the venkat-narasimha/haritha_hospital custom app repo (apps/haritha_hospital/haritha_hospital/fixtures/).

Type	Count	Notes
Custom Fields	78	Employee (PAN, IFSC, approvers), Company (Payroll cost center), Attendance (status extensions), Shift Type (color + HRMS flags), Shift Assignment (Dept link), Holiday List (Telangana regional), Leave Application workflow
Property Setters	189	Largest category — ~120 HRMS, ~50 ERPNext, ~19 Frappe core. Status options, defaults, mandatory toggles, field order
Print Formats	3	Payslip, Shift Card, Leave Application
Notifications	2	Both disabled (Shift assignment change + Leave approval pending)
Letter Heads	2	Haritha Hospitals (default) + Haritha Hospitals — Confidential (HR/Payroll)
Total	274	□ All production-validated on both pberpdev and pberpprod

Quick links

- **HARITHA_HOSPITALS_GUIDE.md** — comprehensive guide (architecture, customizations, migration, ops runbook)
 - **TRACKER.md** — phase-by-phase history + subagent log
 - **docs/phase6/** — Phase 6 Tier 0-8 documentation (foundations, schema, workflow, runbooks, user manuals, testing)
 - **scripts/migrate_master_data.py** — idempotent master data migration (16 DocTypes, 10 gotchas documented)
 - **scripts/recreate_property_setters.py** — idempotent Property Setter recreate (HRMS doesn't list Property Setter as a fixture)
 - **masters/** — 19 source CSVs (canonical reference)
 - **INDEX.md** — curated reading index
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Key operational links

- **Prod:** <https://pberpprod.duckdns.org>
 - **Dev:** <https://pberpdev.duckdns.org>
 - **Venkat VPS prototype:** <https://dev-erp.duckdns.org>
 - **Roster SPA:** <https://pberpprod.duckdns.org/hr/roster>
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Conventions (current)

- **Git commits:** venkat-narasimha <srivenkatnarasimha@gmail.com> (Rule #11)
 - **Custom Fields:** all in haritha_hospital fixtures from day 1 (Rule #9)
 - **HRMS pin:** v16.5.0 only (Lesson #44 — v16.5.1+ breaks on repost_allowed_types)
 - **Shift codes:** 10-char [P][HHMM][S][HHMM] (actual format [GMAN]\d{4}[RS]\d{4} — HRMS-native flags)
 - **Color palette:** G=blue, M=green, A=orange, N=violet (lowercase Tailwind, matching MonthViewTable.vue)
 - **Always restart backend container** after bench install-app (Lesson #153 — gunicorn --preload sys.path freeze)
 - **DB passwords verified monthly** via docker exec erp-\${env}-db-1 printenv MYSQL_ROOT_PASSWORD (Lesson #154)
 - **Holidays:** standard Indian national + 4-5 Telangana regional
 - **QA env skipped** for Haritha — direct dev → prod promotion with custom-app fixtures as the safety net
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Recent milestones

- **2026-08-29 — Phase 6 docs complete** — 22 docs across 8 tiers (docs/phase6/), ~10,000 lines, Mermaid diagrams
- **2026-08-29 — Master data migration prod → dev** — 16 DocTypes via migrate_master_data.py; 8,118 Shift Assignments on dev, all bulk-submitted; idempotent script saved
- **2026-08-29 — P1 outage resolved** — gunicorn --preload sys.path freeze; both envs restarted, zero data loss; LEARNINGS #153
- **2026-08-28 — Custom app haritha_hospital installed** — 274 customizations verified on both envs; idempotent

- **2026-08-28 — Phase 4.10/4.11 Roster crash fixed** — CapitalCase → lowercase Tailwind colors; SPA now renders 211 × 31 cleanly
 - **2026-08-28 — Phase 4.8 Attendance HRMS-recompute** — 6,300 → 9,734 Attendance records; early_exit count 7 → 1,498 (the main bug from Phase 3.8)
 - **2026-08-27 — Phase 3.6/3.7/3.8/3.9/3.10** — bulk-submit 6,314 docs, recreate_property_setters.py, attendance linkage, populate fields, backup bundle fix
 - **2026-08-26 — Phase 3 data import** — 24,511 records across 9 entities
 - **2026-08-25 — Phase 2 site setup** — pberpprod.duckdns.org fresh init + apps installed (frappe, erpnext, hrms 16.5.0, payments)
 - **2026-08-21 — Rollback event** — pberp.duckdns.org env destroyed in Option-B tear-down; preserved CSVs + git history; restart from Phase 1
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References

- **Custom app repo:** https://github.com/venkat-narasimha/haritha_hospital (custom app + 274 fixtures)
 - **Frappe docs:** <https://docs.frappe.io/>
 - **HRMS docs:** <https://docs.frappe.io/hr/>
 - **Frappe framework:** <https://frappeframework.com/docs/>
 - **LEARNINGS:** /root/.openclaw/workspace/.learnings/LEARNINGS.md (workspace-level, includes #44, #106, #114, #142-#150, #151-#157)
 - **MEMORY:** /root/.openclaw/workspace/MEMORY.md (workspace-level tech stack + DB passwords)
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Last updated: 2026-08-31 — refreshed to add Tier 6 (09-compliance) docs + client demo deck + speaker script